THE ISSUE

High Booking Cancellations

Unpredictable cancellations lead to revenue loss, inefficient operations, and inaccurate demand forecasting

THE CONSEQUENCES

Revenue Uncertainty and Financial Loss

High cancellation rates inflate projected earnings, leaving rooms empty and revenue unfulfilled

Operational Disruption & Inefficiency

Frequent cancellations cause wasted prep time, poor resource allocation, and lower occupancy rates

OUR GOAL

Predict booking cancellations

based on customer characteristics and booking details

to help hotels

make smarter operational and pricing decisions

improving

profitability and guest experience

SOLVING THRU DATA SCIENCE

Data Source

Hotel Booking Data kaggle

Two hotels in Portugal (City & Resort) from July 2015 to August 2017

Data Dimensions

119,390 rows (bookings)
32 columns (features)

Machine Learning

Classification/ Clustering

logistic regression, random forest, decision tree, clustering, etc

Hypothesis

Ho: cancellations occur randomly
Ha: significant predictive powers

Key Features

Target

booking status
cancelled (1)
not cancelled (0)

Predictors

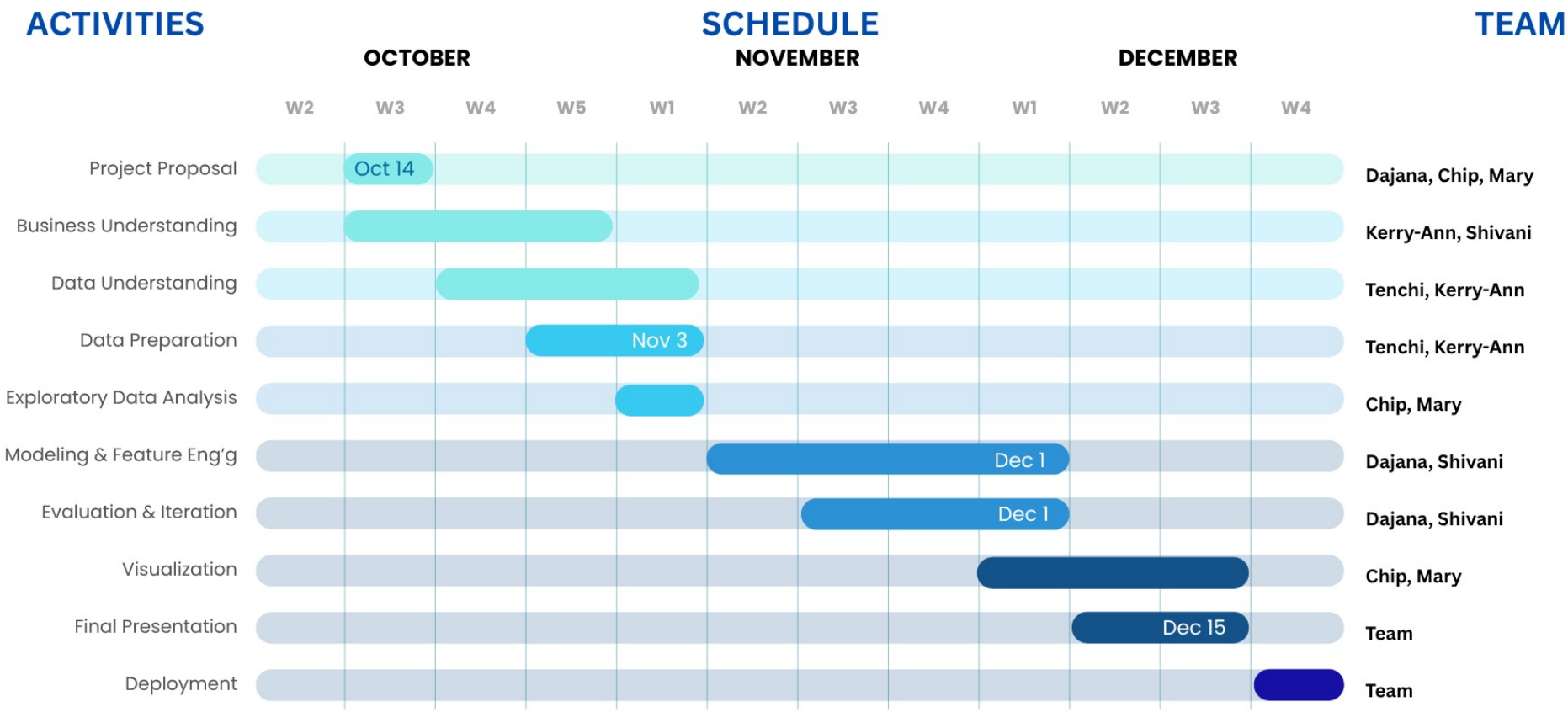
booking attributes
lead time
arrival date
length of stay
reservation status
average daily rate

guest information
# of adults
# of children
country of origin

booking behavior
previous cancellation
repeated guest

hotel/room details
hotel type
reserved vs assigned
deposit type

PLAN


ACTIVITIES
SCHEDULE
TEAM
OCTOBER
NOVEMBER
DECEMBER
W2
W3
W4
W5
W1
W2
W3
W4
W1
W2
W3
W4
Project Proposal
Oct 14
Dajana, Chip, Mary
Business Understanding
Kerry-Ann, Shivani
Data Understanding
Tenchi, Kerry-Ann
Data Preparation
Nov 3
Tenchi, Kerry-Ann
Exploratory Data Analysis
Chip, Mary
Modeling & Feature Eng'g
Dec 1
Dajana, Shivani
Evaluation & Iteration
Dec 1
Dajana, Shivani
Visualization
Chip, Mary
Final Presentation
Dec 15
Team
Deployment
Team

ASSUMPTIONS

Data still representative after pre-processing
Minimal missing or erroneous values after preprocessing

Historical booking data is accurate and reflective of actual events
Data we collected is valid and is recorded as it happens

Data captures typical conditions
Data reflect typical booking behavior and is consistently collected across time and properties.

Readers have domain knowledge and will understand report
Have the capacity to understand mode;

RISKS

Handling Missing & Inconsistent Data
Data cleaning, outlier handling, missing values imputing (dropping vs filling) to be done

Model Overfitting
Continuous Cross Validation

Overlooked/ Ommited Variables
like marketing campaigns, competition. Model-bounded decisions

Misinterpretation & Misuse
Disclaimers on correlation interpreted as causation. Careful interpretation guide

Scope Creep
Explaining prediction scope of booking cancellation; excludes demand forecasting

BUSINESS BENEFITS

Better demand forecasting
forecasts are closer to true demand.
Model can automatically flag cancellers

Improve Revenue & Reduce Avoidable Loss
Identifying high-risk bookings early allow hotels take preventive actions which affects revenues

Optimize staffing & resource planning
Accurate booking forecasts allow efficient staffing, housekeeping, and supply management.

Better safeguards and offers
Booking deposits and cancellation window times for cancellers or offers to prevent such

Better customer segment targeting
understanding true guests helps Identify customer segments for targetting

Accomodate true customers and develop loyalty
Allocate rooms to true predicted customers and improve customer experience

insightout

Decoding Data from Insights to Impact